Adv ML for Gen-AI

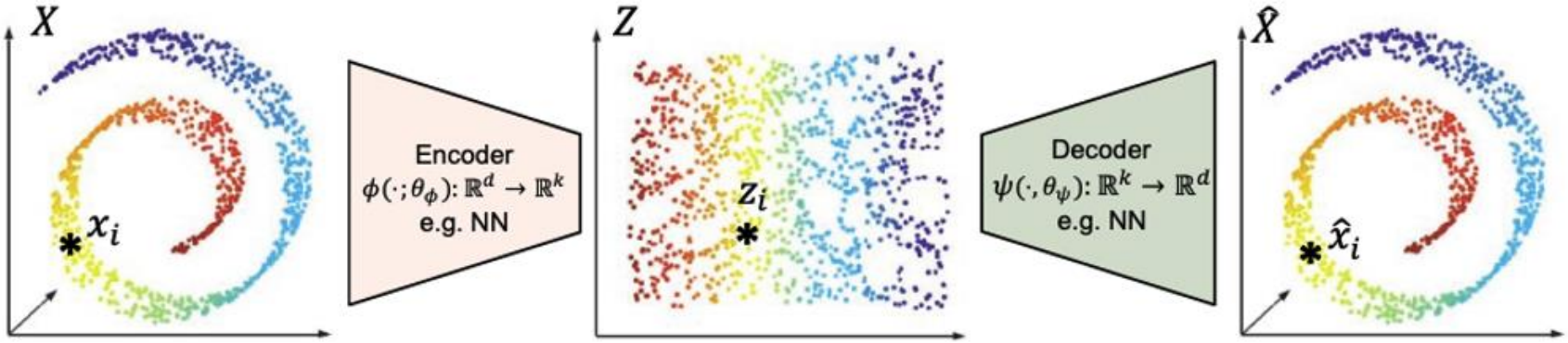
Generative Adversarial Network

https://ml-graph.github.io/spring-2026/

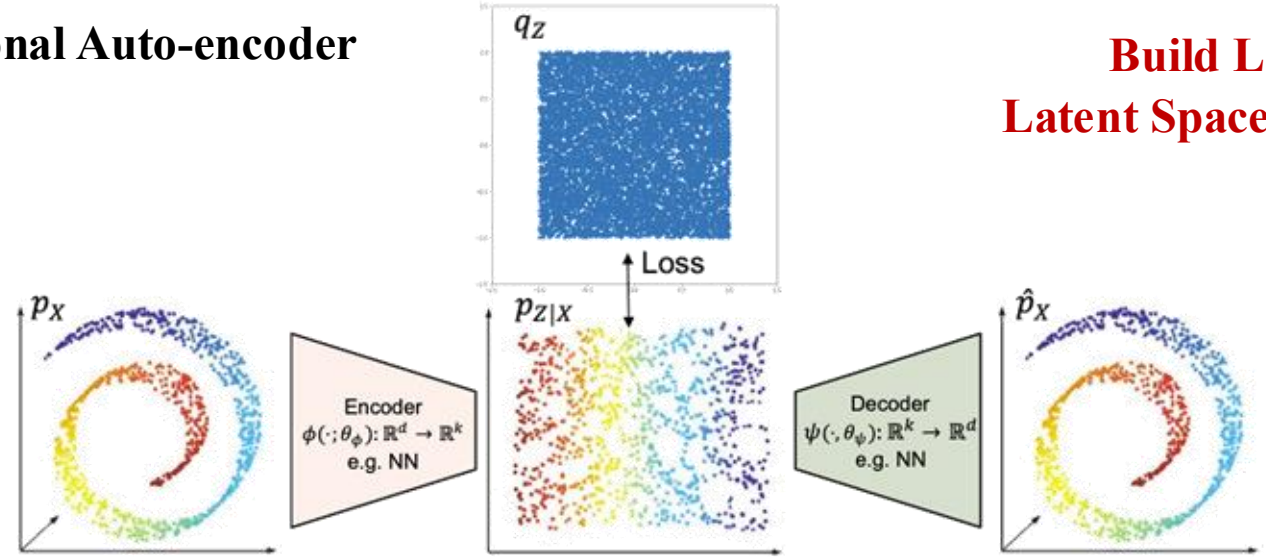
Yu Wang, Ph.D.
Assistant Professor
Department of Computer Science
University of Oregon
Personal: https://yuwang0103.github.io/
Lab: https://kindlab-fly.github.io/

Recap

Auto-encoder based generative modeling



Variational Auto-encoder

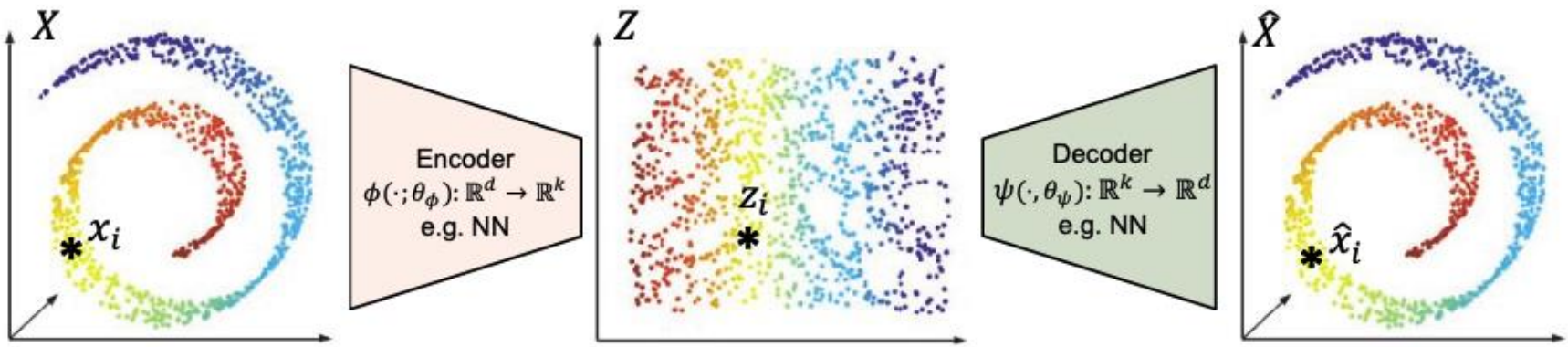


Build Latent Space

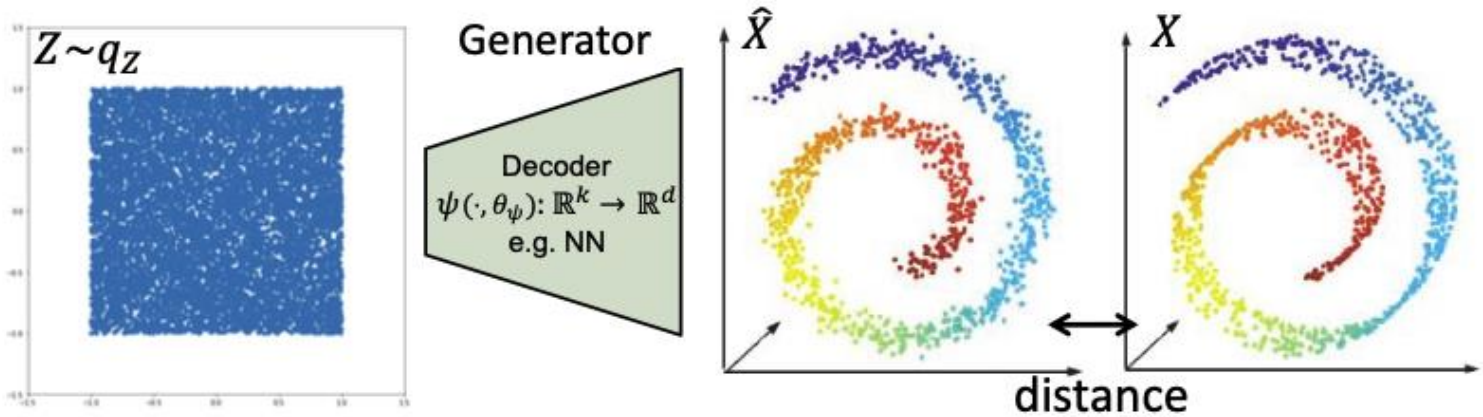
Latent Space – Decode Image

Question

Auto-encoder based generative modeling



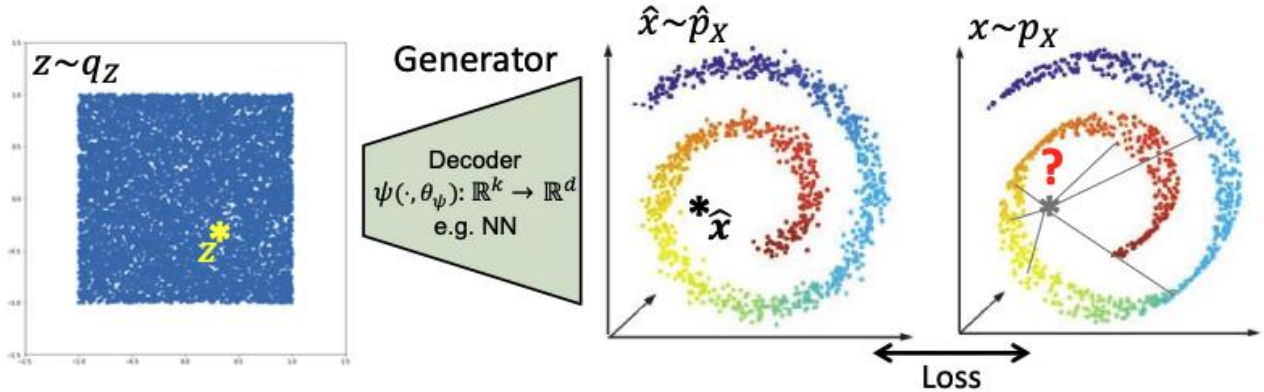
Encoder-less Generative Modeling



Can we remove encoder, but directly start from latent space and map?

Question

Generative Adversarial Networks: Main Idea



We sample $z \sim q_Z$ and generate $\hat{x} = \psi(z; \theta_\psi)$. Note that we want $\hat{x}$ to sit on the manifold of the original data, but we do not know the corresponding point for $\hat{x}$ in X!

How to compute the likelihood?

Sample-wise distances won't work in this setting, and one must rely on distribution-based distances.

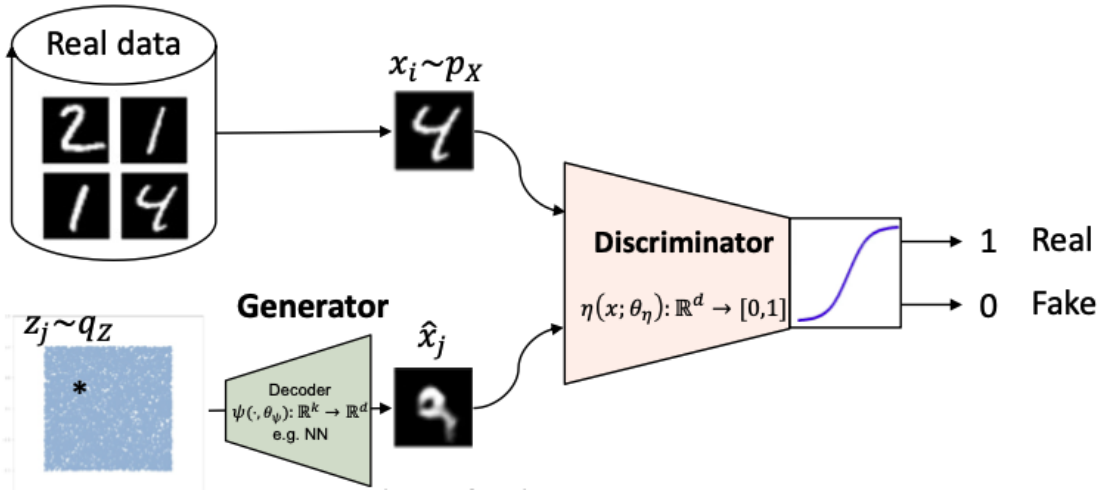
Can we train a neural network to tell the difference and try to minimize that difference?

Generative Adversarial Networks

Generative Adversarial Networks (GAN)

The idea in GANs is to use an adversarial network that tries to distinguish the points on the manifold of the data from any other points in $\mathbb{R}^d$. Let $\eta(x;\theta_\eta): \mathbb{R}^d \to [0,1]$ be the discriminator:

$$\min_{\theta_\psi} \max_{\theta_\eta} \frac{1}{N}\sum_i \log(\eta(x_i;\theta_\eta)) + \frac{1}{M}\sum_j \log\left(1-\eta(\hat{x}_j;\theta_\eta)\right), \qquad where\ \hat{x}_j = \psi(z_j;\theta_\psi)$$



Generative Adversarial Networks

Minimax objective

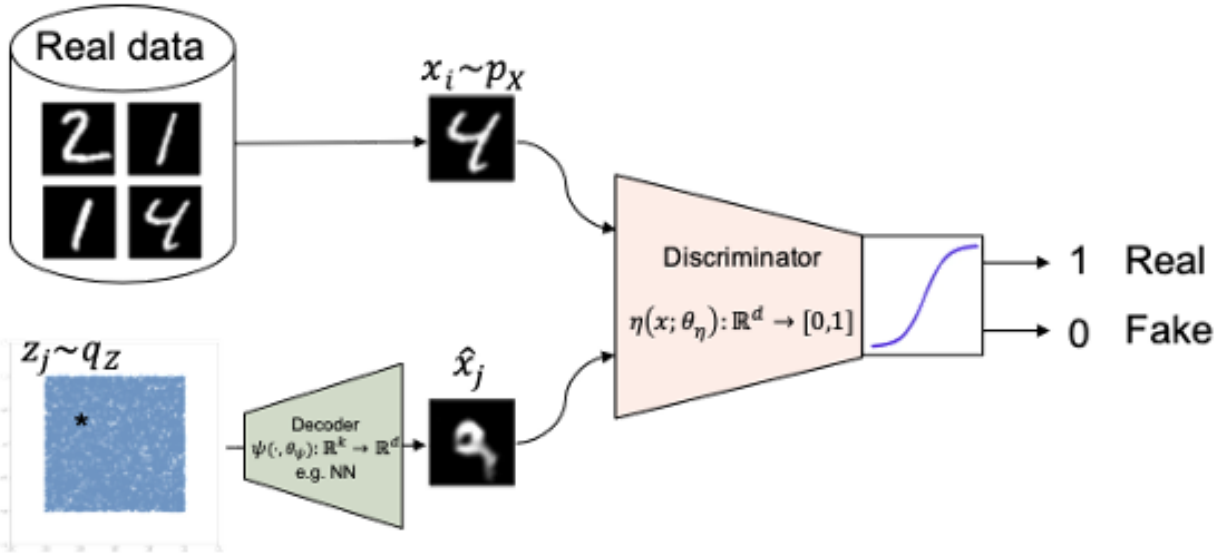
A GAN has two networks:

- **Generator** $G_\theta : \mathbb{R}^d \to \mathbb{R}^{784}$ — maps noise $z \sim \mathcal{N}(0, I)$ to fake images
- **Discriminator** $D_\phi : \mathbb{R}^{784} \to [0, 1]$ — predicts probability that an image is real

They play a **minimax game**:

$$\min_G \max_D \; V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))]$$

- D wants to output 1 for real, 0 for fake → **maximise** V
- G wants D to output 1 for its fakes → **minimise** V (equivalently, maximise $\log D(G(z))$)

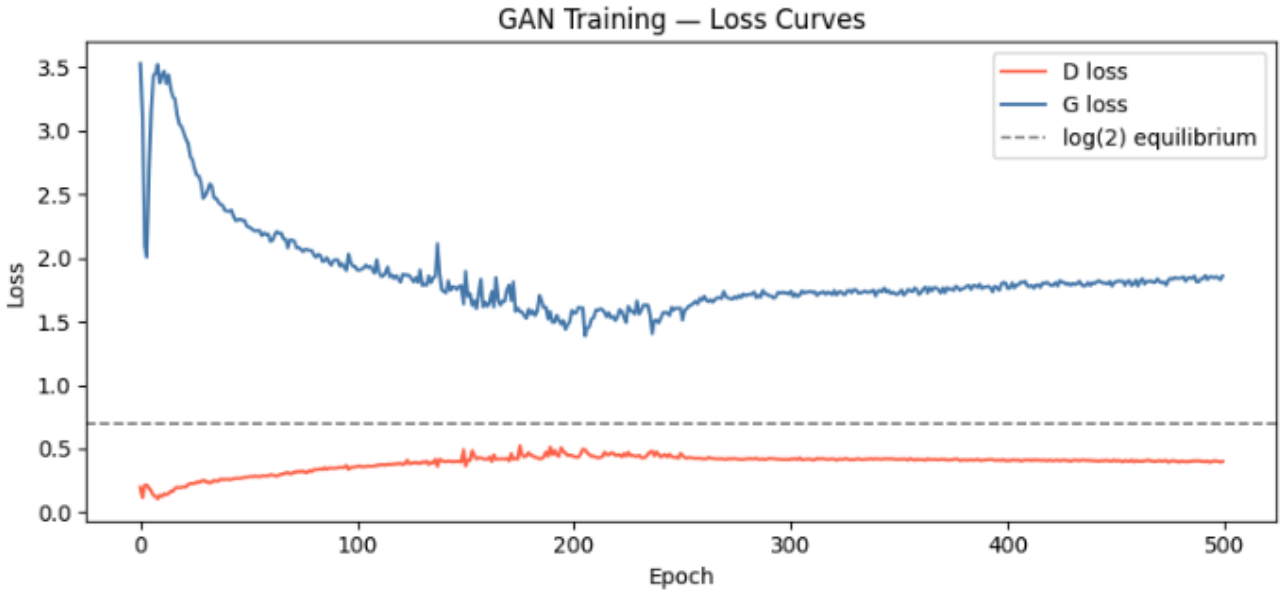


Generative Adversarial Networks

```python
class Generator(nn.Module):
    def __init__(self, latent_dim=LATENT_DIM):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(latent_dim, 256),
            nn.BatchNorm1d(256),
            nn.LeakyReLU(0.2),
            nn.Linear(256, 512),
            nn.BatchNorm1d(512),
            nn.LeakyReLU(0.2),
            nn.Linear(512, 784),
            nn.Sigmoid(),           # output in [0
        )
    def forward(self, z):
        return self.net(z)


class Discriminator(nn.Module):
    def __init__(self):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(784, 512),
            nn.LeakyReLU(0.2),
            nn.Dropout(0.3),
            nn.Linear(512, 256),
            nn.LeakyReLU(0.2),
            nn.Dropout(0.3),
            nn.Linear(256, 1),
            nn.Sigmoid(),
        )
    def forward(self, x):
        return self.net(x)
```

```python
# Fixed noise for visualisation — same z throughout training
fixed_z = torch.randn(16, LATENT_DIM, device=device)

for epoch in range(1, N_EPOCHS + 1):
    epoch_d, epoch_g = 0.0, 0.0
    for real_imgs, _ in train_dl:
        real_imgs = real_imgs.to(device)
        B = real_imgs.size(0)

        real_labels = torch.ones (B, 1, device=device)
        fake_labels = torch.zeros(B, 1, device=device)

        # — Train Discriminator ————————————————————————
        z         = torch.randn(B, LATENT_DIM, device=device)
        fake_imgs = G(z).detach()          # detach: don't backprop into G yet

        loss_D_real = criterion(D(real_imgs), real_labels)
        loss_D_fake = criterion(D(fake_imgs), fake_labels)
        loss_D      = (loss_D_real + loss_D_fake) / 2

        opt_D.zero_grad(); loss_D.backward(); opt_D.step()

        # — Train Generator ————————————————————————
        z         = torch.randn(B, LATENT_DIM, device=device)
        fake_imgs = G(z)

        # Non-saturating loss: G maximises log D(G(z))
        loss_G = criterion(D(fake_imgs), real_labels)

        opt_G.zero_grad(); loss_G.backward(); opt_G.step()

        epoch_d += loss_D.item()
        epoch_g += loss_G.item()

    d_losses.append(epoch_d / len(train_dl))
    g_losses.append(epoch_g / len(train_dl))

    if epoch % 5 == 0:
        print(f'Epoch {epoch:3d}/{N_EPOCHS}  D_loss={d_losses[-1]:.4f}  G_loss={g_losses[-1]:.4f}')

print('Training complete.')
```

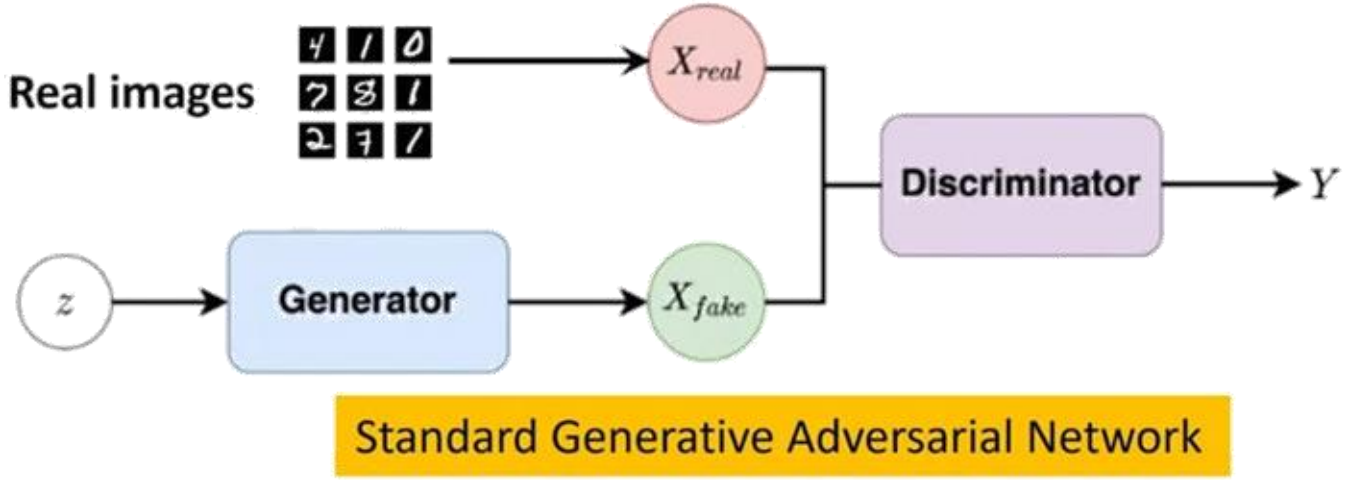
Generative Adversarial Networks

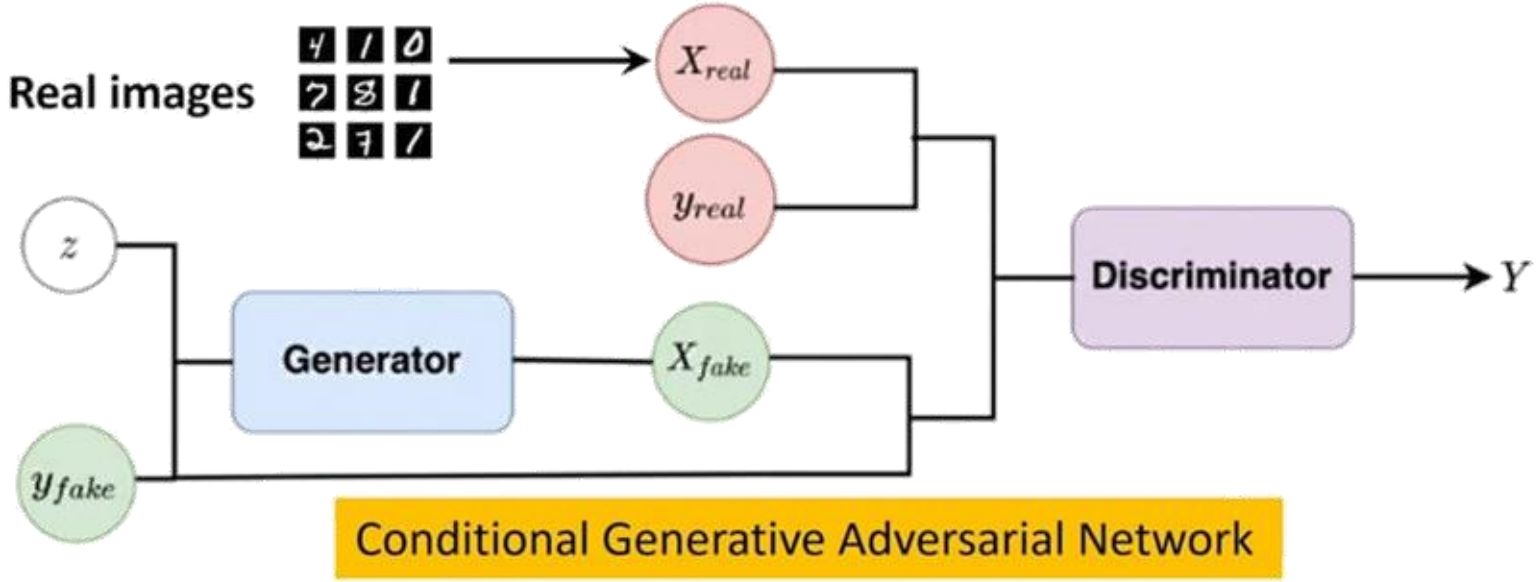


Discriminator cannot recognize what is real and what is fake

Generator continuously confuses discriminator

Class Conditional GAN





Class Conditional GAN

With a vanilla GAN, $G(z)$ generates a random digit — we can't ask for a specific class.

Conditional GAN (Mirza & Osindero, 2014) fixes this by conditioning both networks on a label y:

$$\min_G \max_D \; V(D,G) = \mathbb{E}_{x,y}[\log D(x \mid y)] + \mathbb{E}_{z,y}[\log(1 - D(G(z \mid y) \mid y))]$$

Both G and D receive the class label as an additional input.

At test time, we can choose any label y to get a digit of that class.

The standard approach: **embedding + concatenation**

```
Generator:     [z (64-d) | embed(y) (16-d)]  →  784
Discriminator: [x (784)  | embed(y) (16-d)]  →  1
```

Class Conditional GAN

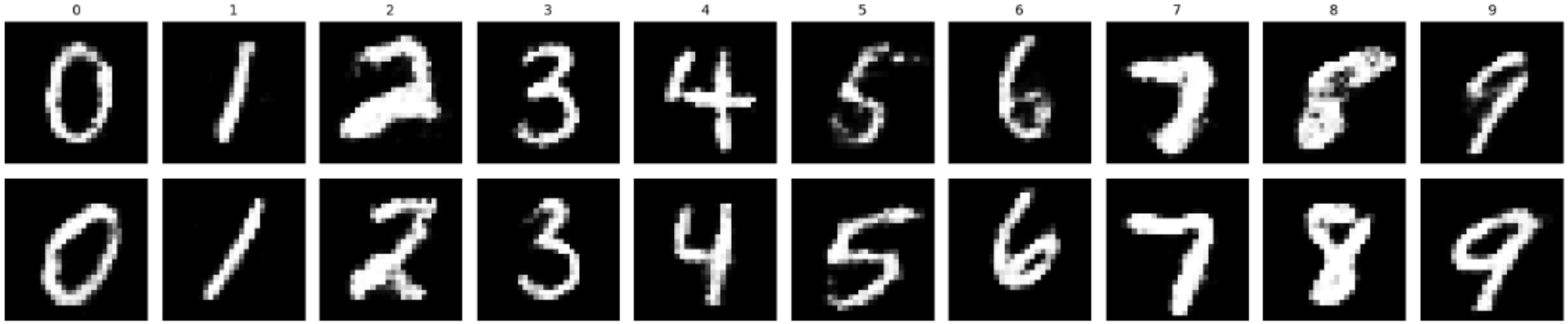
```python
class CondGenerator(nn.Module):
    def __init__(self, latent_dim=LATENT_DIM, embed_dim=EMBED_DIM, n_classes=N_CLASSES):
        super().__init__()
        self.label_emb = nn.Embedding(n_classes, embed_dim)
        self.net = nn.Sequential(
            nn.Linear(latent_dim + embed_dim, 256),
            nn.BatchNorm1d(256),
            nn.LeakyReLU(0.2),
            nn.Linear(256, 512),
            nn.BatchNorm1d(512),
            nn.LeakyReLU(0.2),
            nn.Linear(512, 784),
            nn.Sigmoid(),
        )
    def forward(self, z, y):
        emb = self.label_emb(y)          # (B, embed_dim)
        inp = torch.cat([z, emb], dim=1) # (B, latent_dim + embed_dim)
        return self.net(inp)


class CondDiscriminator(nn.Module):
    def __init__(self, embed_dim=EMBED_DIM, n_classes=N_CLASSES):
        super().__init__()
        self.label_emb = nn.Embedding(n_classes, embed_dim)
        self.net = nn.Sequential(
            nn.Linear(784 + embed_dim, 512),
            nn.LeakyReLU(0.2),
            nn.Dropout(0.3),
            nn.Linear(512, 256),
            nn.LeakyReLU(0.2),
            nn.Dropout(0.3),
            nn.Linear(256, 1),
            nn.Sigmoid(),
        )
    def forward(self, x, y):
        emb = self.label_emb(y)          # (B, embed_dim)
        inp = torch.cat([x, emb], dim=1) # (B, 784 + embed_dim)
        return self.net(inp)
```

```python
for epoch in range(1, N_EPOCHS + 1):
    epoch_d, epoch_g = 0.0, 0.0
    for real_imgs, labels in train_dl:
        real_imgs = real_imgs.to(device)
        labels    = labels.to(device)
        B = real_imgs.size(0)

        real_labels = torch.ones (B, 1, device=device)
        fake_labels = torch.zeros(B, 1, device=device)

        # — Train cD ————————————————————————————————————————
        z         = torch.randn(B, LATENT_DIM, device=device)
        fake_imgs = cG(z, labels).detach()

        loss_D = (criterion(cD(real_imgs, labels), real_labels) +
                  criterion(cD(fake_imgs, labels), fake_labels)) / 2

        opt_cD.zero_grad(); loss_D.backward(); opt_cD.step()

        # — Train cG ————————————————————————————————————————
        z         = torch.randn(B, LATENT_DIM, device=device)
        fake_imgs = cG(z, labels)
        loss_G    = criterion(cD(fake_imgs, labels), real_labels)

        opt_cG.zero_grad(); loss_G.backward(); opt_cG.step()

        epoch_d += loss_D.item()
        epoch_g += loss_G.item()
```

Class Conditional GAN

```python
# — Show class-conditioned generation ————————————————————————————
cG.eval()
with torch.no_grad():
    cgan_samples = cG(fixed_z_cgan, fixed_y_cgan).cpu().numpy().reshape(-1, 28, 28)
cG.train()

fig, axes = plt.subplots(2, 10, figsize=(16, 4))
for i in range(2):
    for j in range(10):
        axes[i, j].imshow(cgan_samples[i*10+j], cmap='gray', vmin=0, vmax=1)
        axes[i, j].axis('off')
        if i == 0:
            axes[i, j].set_title(str(j), fontsize=10)
fig.suptitle('cGAN: each column is conditioned on digit label 0–9', fontsize=12)
plt.tight_layout()
plt.show()
```

```python
# Fixed z + labels 0–9 twice for a 2×10 display grid
fixed_z_cgan  = torch.randn(20, LATENT_DIM, device=device)
fixed_y_cgan  = torch.tensor(list(range(10)) * 2, device=device)
```



QA